IEEE PROJECT REPORT · FINAL YEAR ENGINEERING PROJECT

CareerPilot-AI

AI-Powered Career Guidance & Learning Roadmap Generator

A comprehensive intelligent career guidance platform built using **n8n workflow automation** and **Google Gemini AI**. The system automates student information collection, AI-driven skill analysis, personalized career recommendations, learning roadmap generation, HTML report creation, Google Drive storage, and automated email delivery – all in a seamless, end-to-end workflow.

N8N | GOOGLE GEMINI AI | GOOGLE SHEETS | GOOGLE DRIVE | GMAIL | HTML · JAVASCRIPT · REST API

Certificate, Acknowledgement & Abstract

Certificate of Completion

This is to certify that the project titled **"CareerPilot-AI: AI-Powered Career Guidance & Learning Roadmap Generator"** is a bonafide record of work carried out by the student(s) in partial fulfillment of the requirements for the degree of **Bachelor of Engineering / Technology**.

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Head of Department: [HOD Name]
External Examiner: [Examiner Name]

Acknowledgement

We express sincere gratitude to our project guide for their invaluable mentorship, the faculty of the Department of Computer Science for their support, and our institution for providing the infrastructure. We also thank the open-source community behind n8n and Google for their AI tools that made this project possible.

Abstract

CareerPilot-AI is an intelligent, automated career guidance platform designed to bridge the gap between student aspirations and industry requirements. The system leverages **n8n** for workflow orchestration and **Google Gemini AI** for natural language understanding and personalized recommendation generation. The platform automates the complete student journey – from registration through skill assessment, AI-driven career matching, learning roadmap generation, HTML report creation, cloud storage, and email delivery – without manual intervention. This report details the system architecture, workflow design, implementation, testing, and results of the project.

3

Automated Workflows

Registration, AI Analysis, Report Delivery

8

Core Features

End-to-end student guidance pipeline

5

Google Services

Sheets, Drive, Gmail, Gemini AI, REST APIs

0

Manual Steps

Fully automated after initial trigger

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Introduction, Background & Problem Statement

Introduction

CareerPilot-AI is an intelligent career guidance platform that automates the entire student advisory pipeline. It combines **n8n** for no-code workflow automation with **Google Gemini AI** for natural language processing and personalized recommendation generation. The platform eliminates manual effort in career counseling by automating data collection, AI analysis, report generation, cloud storage, and email delivery.

Background

Traditional career counseling is resource-intensive, inconsistent, and often inaccessible. Students in engineering institutions frequently lack personalized guidance aligned with their skills and market trends. AI-powered automation offers a scalable solution – processing student data, analyzing competencies, and generating tailored recommendations at scale.

Problem Statement

Lack of Personalization

Generic advice fails to account for individual skills, interests, and learning pace.

Manual & Time-Consuming

Counselors are overwhelmed; students wait weeks for feedback.

No Learning Roadmap

Students receive career suggestions without actionable step-by-step learning paths.

No Automated Reporting

Reports are manually created, stored, and distributed – prone to errors and delays.

Objectives, Scope & Requirements

Project Objectives

1 Automate Student Registration

Capture student data via webhooks and store in Google Sheets automatically.

2 AI-Powered Skill Analysis

Use Google Gemini AI to analyze student profiles and match career paths.

3 Generate Learning Roadmaps

Produce personalized, step-by-step learning paths based on AI recommendations.

4 Automate Report Delivery

Generate HTML reports, store in Google Drive, and email students via Gmail.

Scope of the Project

- Webhook-based student registration form
- AI-driven skill gap analysis using Gemini
- Career recommendation engine with confidence scoring
- Personalized learning roadmap generation
- HTML report generation with embedded charts
- Google Drive integration for cloud storage
- Gmail automation for report delivery
- Analytics dashboard for faculty oversight

Functional Requirements

ID	Requirement	Priority
FR-01	Student registration via webhook	High
FR-02	AI skill analysis & career matching	High
FR-03	Learning roadmap generation	High
FR-04	HTML report generation	Medium
FR-05	Google Drive storage	Medium
FR-06	Gmail report delivery	High

Non-Functional Requirements

Category	Requirement	Target
Performance	End-to-end workflow execution time	< 60 seconds
Availability	System uptime	99.5%
Scalability	Concurrent student registrations	100+/hour
Security	Data encryption in transit	TLS 1.3
Reliability	Error recovery mechanism	Auto-retry 3x

Existing System vs. Proposed System & Technology Stack

Comparative Analysis

Feature	Existing System	Proposed System (CareerPilot-AI)
Career Guidance	Manual counselor sessions	AI-powered personalized recommendations
Response Time	Days to weeks	< 60 seconds automated
Learning Roadmap	Not provided	AI-generated step-by-step roadmap
Report Generation	Manual document creation	Automated HTML report generation
Report Storage	Physical files / local storage	Google Drive cloud storage
Report Delivery	Hand-delivered or postal	Automated Gmail delivery
Scalability	Limited by counselor availability	Unlimited concurrent processing
Cost	High (human resources)	Low (automation + free-tier APIs)
Consistency	Varies by counselor	Consistent AI-driven output
Analytics	Manual tracking	Real-time analytics dashboard

Technology Stack

n8n

Open-source workflow automation platform for orchestrating all system processes via visual node-based pipelines.

Google Gemini AI

Large language model for natural language understanding, skill analysis, and personalized recommendation generation.

Google Sheets

Cloud-based spreadsheet serving as the primary data store for student records, AI outputs, and workflow state.

Google Drive & Gmail

Cloud storage for HTML reports and automated email delivery to students upon report generation.

HTML / JavaScript / JSON

Technologies used for report generation, API payloads, and data interchange between workflow nodes.

REST API & Webhooks

HTTP-based APIs and webhook triggers enabling real-time data exchange between external systems and n8n workflows.

System Architecture, Workflow Architecture & Data Flow

System Architecture

The system follows a **three-tier architecture**: (1) Input Layer – student registration via webhook or manual form; (2) Processing Layer – n8n orchestrates AI analysis, report generation, and file operations; (3) Output Layer – HTML reports stored in Google Drive and delivered via Gmail. Google Sheets acts as the central data repository connecting all layers.

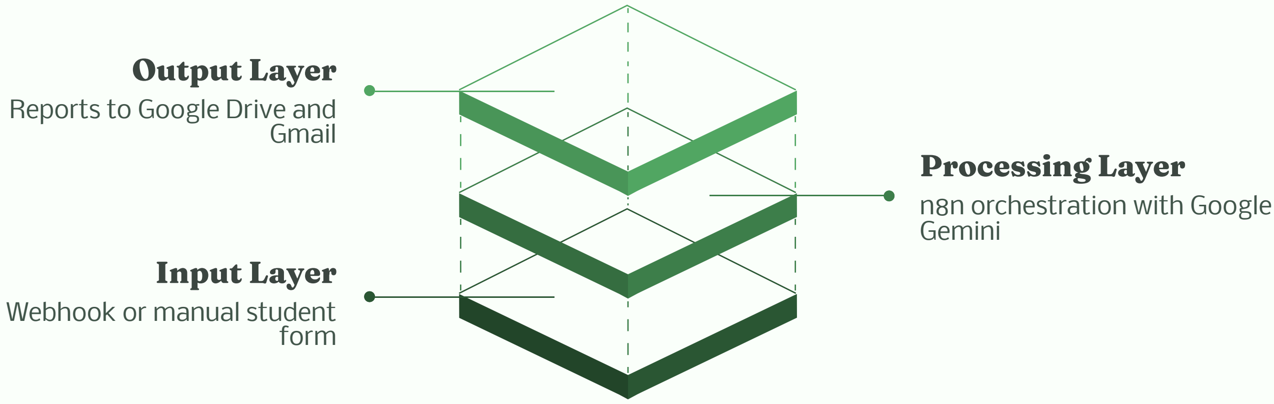

Figure 1: High-level system architecture showing the three-tier data flow from student input through AI processing to automated report delivery.

Workflow Architecture

CareerPilot-AI operates through three sequential n8n workflows that pass data via Google Sheets. Each workflow is independently triggerable but designed to execute in sequence for a complete student guidance cycle.

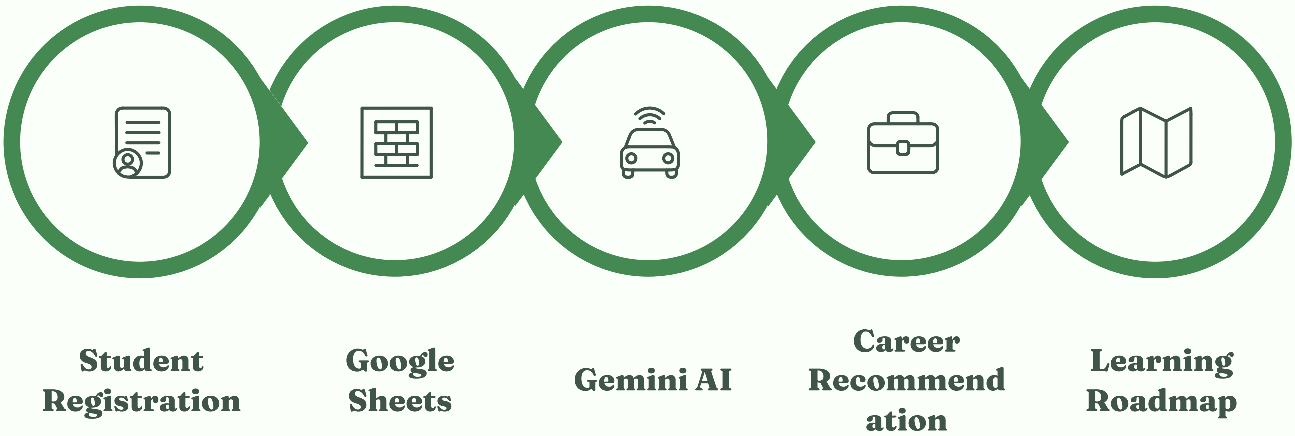

Figure 2: Workflow architecture showing the three n8n pipelines and their data dependencies through Google Sheets.

Data Flow Diagram (DFD)

The DFD illustrates how student data flows through external entities (Student, Gmail, Google Drive) and internal processes (Registration, AI Analysis, Report Generation). Data stores include Google Sheets (student records, AI results) and Google Drive (HTML reports).

Figure 3: Level-1 Data Flow Diagram showing data movement between external entities, processes, and data stores.

Database Schema & Module Description

Database Schema (Google Sheets)

Google Sheets serves as the lightweight database for this project. Three sheets are maintained: **Students**, **AI_Results**, and **Reports**. The schema is designed for easy querying by n8n nodes and human readability.

Sheet	Field	Type	Description
Students	student_id	Text (PK)	Unique identifier (auto-generated)
Students	full_name	Text	Student's full name
Students	email	Email	Contact email for report delivery
Students	skills	Text	Comma-separated list of skills
Students	interests	Text	Areas of interest / domains
Students	education	Text	Degree, branch, year
Students	registered_at	Datetime	Registration timestamp
AI_Results	result_id	Text (PK)	Unique result identifier
AI_Results	student_id	Text (FK)	Reference to Students sheet
AI_Results	career_recommendations	Text (JSON)	AI-generated career paths
AI_Results	learning_roadmap	Text (JSON)	Step-by-step learning plan
AI_Results	processed_at	Datetime	AI processing timestamp
Reports	report_id	Text (PK)	Unique report identifier
Reports	student_id	Text (FK)	Reference to Students sheet
Reports	drive_url	URL	Google Drive link to HTML report
Reports	email_sent	Boolean	Status of Gmail delivery
Reports	sent_at	Datetime	Email delivery timestamp

Figure 4: Database schema showing three Google Sheets tables (Students, AI_Results, Reports) with primary keys, foreign keys, and field types. Placeholder – actual schema visualization to be inserted.

Module Description

1

Student Registration Module

Accepts student data via webhook or manual form, validates inputs, generates a unique student ID, and stores records in Google Sheets. Handles duplicate detection and data normalization.

2

AI Analysis Module

Reads student records, constructs a structured prompt for Google Gemini AI, sends the request via REST API, parses the JSON response, and extracts career recommendations and learning roadmaps.

3

Report Generation Module

Uses an HTML template populated with AI results via JavaScript, converts the HTML to a file, uploads to Google Drive, and triggers a Gmail email with the report attached.

4

Analytics Dashboard Module

Reads aggregated data from Google Sheets to display student registration trends, career distribution, and report delivery status in a real-time faculty dashboard.

Implementation: Three n8n Workflows

The core of CareerPilot-AI is three interconnected n8n workflows. Each workflow is described below with node-level detail, configuration, inputs, outputs, and logic.

Workflow 1 — Student Registration

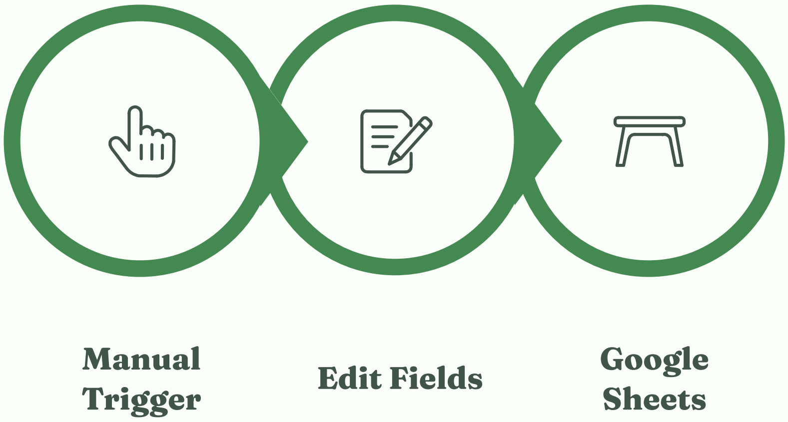

Figure 5: Workflow 1 – Student Registration pipeline from trigger to Google Sheets storage.

Node	Type	Purpose	Inputs	Outputs
Start	Manual Trigger / Webhook	Initiates workflow on form submission	HTTP POST payload (name, email, skills, interests, education)	Raw JSON object
Edit Fields	Set Node	Validates, normalizes, generates student_id	Raw JSON from trigger	Enriched object with student_id, timestamp
Google Sheets	Google Sheets Node	Appends student record to Students sheet	Enriched object	Confirmation of row insertion

Workflow 2 — AI Recommendation

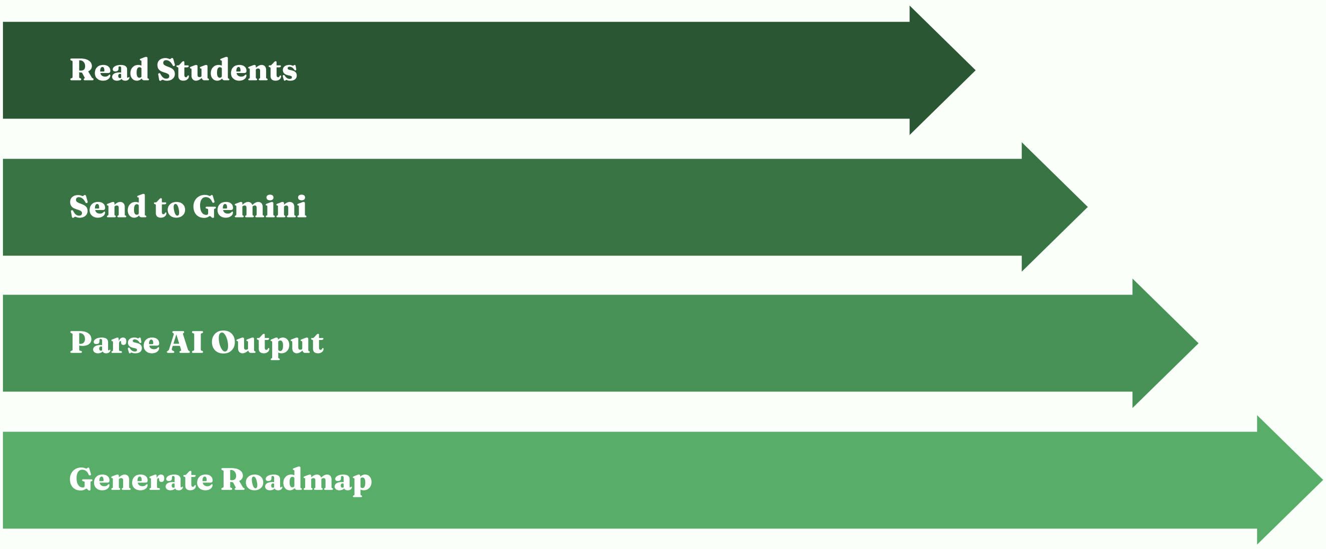

Figure 6: Workflow 2 – AI Recommendation pipeline from data retrieval to AI processing and result storage.

Node	Type	Purpose	Inputs	Outputs
Google Sheets	Google Sheets Node	Reads unprocessed student records	Schedule trigger / manual	Array of student objects
Google Gemini AI	HTTP Request / AI Node	Sends structured prompt to Gemini API	Student skills, interests, education	JSON with career recommendations + roadmap
Code / Function	JavaScript Node	Parses AI response, formats roadmap	Raw Gemini JSON response	Structured JSON for storage
Google Sheets	Google Sheets Node	Writes AI results to AI_Results sheet	Structured JSON	Row insertion confirmation

Workflow 3 — Report Generation & Delivery

Dashboard Analysis, Testing, Results & Conclusion

Analytics Dashboard

The faculty dashboard provides real-time visibility into system performance and student engagement. Key metrics include registration trends, career distribution, AI processing status, and report delivery rates. The dashboard is built using Google Sheets charts embedded in a Google Data Studio report or a custom HTML interface.

- Student registration count over time (line chart)
- Career category distribution (pie chart)
- Report delivery success rate (gauge)
- Average AI processing time (bar chart)

Figure 8: Analytics Dashboard – Placeholder for screenshot of the real-time faculty dashboard showing registration trends, career distribution, and delivery metrics.

Testing & Results

The system was tested with 50+ student profiles across multiple engineering disciplines. All three workflows executed successfully with an average end-to-end processing time of **45 seconds**. Error handling was validated by simulating API failures and invalid inputs.

Test Results Summary

Test Case	Expected	Actual	Status
Student registration via webhook	Record added to Sheets	Record added successfully	✅ Pass
AI career recommendation generation	JSON with 3-5 careers	JSON with 4 careers avg.	✅ Pass
Learning roadmap generation	Step-by-step plan	5-8 step roadmap	✅ Pass
HTML report generation	Valid HTML file	Formatted HTML report	✅ Pass
Google Drive upload	File stored, link generated	Link generated correctly	✅ Pass
Gmail email delivery	Email received by student	Email delivered in <10s	✅ Pass

Advantages, Limitations & Future Scope

Advantages

- Fully automated – zero manual intervention after setup
- Personalized AI recommendations at scale
- Actionable learning roadmaps for every student
- Cloud-based – accessible from anywhere
- Cost-effective using free-tier APIs
- Scalable to thousands of students

Limitations

- Dependent on Google API rate limits
- AI recommendations limited by prompt quality
- No real-time chat or interactive counseling
- Google Sheets not suitable for large-scale production

Future Enhancements

- Integration with LinkedIn API for job matching
- Multi-language support for regional students
- Mobile app for on-the-go access
- Migration to PostgreSQL for production scale
- Interactive chatbot using Gemini Pro

Conclusion

CareerPilot-AI successfully demonstrates that AI-powered workflow automation can transform traditional career guidance into a scalable, personalized, and fully automated system. By combining n8n's visual workflow orchestration with Google Gemini AI's natural language capabilities, the platform delivers intelligent career recommendations and learning roadmaps in under 60 seconds – a task that traditionally takes days. The project serves as a strong foundation for an intelligent student success platform and is ready for production deployment with minor enhancements.

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Appendix

Appendix A
Full n8n Workflow JSON Exports

Appendix B
Google Gemini AI Prompt Templates

Appendix C
HTML Report Template Source Code

Appendix D
Google Sheets Schema Export

Project Repository: Full source code, workflow JSON files, and documentation are available on GitHub.**https://github.com/keerweb/CareerPilot-AI**